

PANJ, Tajikistan

WMO: 389470

Lat: 37.233N

Lon: 69.083E

Elev: 327

StdP: 97.46

Time Zone: 5.00 (E05)

Period: 83-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.5	-4.3	-10.6	1.6	-8.0	-6.3	2.3	-3.2	8.7	5.8	5.1	4.2	1.4	170	0.384

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.3	39.1	24.3	38.1	23.3	36.9	22.6	25.6	36.9	24.6	36.0	23.7	34.9	1.3	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.1	17.4	34.5	20.7	16.0	33.1	19.6	14.9	32.0	81.4	36.9	76.6	36.2	72.8	35.1	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.2	3.7	2.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.4	4.8	6.2	12.0	18.2	24.0	28.4	29.9	27.7	23.9	17.2	10.3	5.8
	DBStd	9.62	3.69	5.12	4.03	4.04	3.56	1.78	1.60	1.83	1.98	3.72	4.28	4.07
	HDD10.0	490	161	120	27	1	1	0	0	0	0	2	43	136
	HDD18.3	1709	418	340	198	50	7	0	0	0	1	65	242	388
	CDD10.0	3201	2	13	88	248	434	552	617	548	416	224	52	6
	CDD18.3	1379	0	0	2	47	183	302	358	290	167	29	1	0
	CDH23.3	17699	0	2	34	453	1995	4033	5099	3758	1932	379	14	0
	CDH26.7	9889	0	0	4	141	955	2371	3147	2201	955	114	2	0
Wind	WSAvg	1.3	1.2	1.3	1.5	1.4	1.5	1.4	1.2	1.1	1.2	1.2	1.2	1.2
Precipitation	PrecAvg	482	57	83	84	69	45	15	15	13	8	16	36	40
	PrecMax	634	143	169	148	105	129	47	42	26	30	44	107	86
	PrecMin	270	21	32	14	22	6	2	2	2	2	1	7	8
	PrecStd	90	27	33	32	22	26	10	9	6	7	12	24	21
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.3	22.4	27.0	32.6	37.1	39.8	40.8	39.1	35.8	32.1	25.3	18.7
		MCWB	10.4	13.8	17.5	19.8	21.3	23.6	25.5	24.6	21.6	19.7	14.3	10.9
	2%	DB	14.1	18.8	23.5	30.0	35.2	38.3	39.6	37.7	34.7	29.6	22.0	16.1
		MCWB	9.2	12.0	15.7	19.6	21.0	22.6	24.8	23.6	20.9	18.3	13.3	10.3
	5%	DB	12.4	15.9	21.1	28.1	33.7	37.1	38.7	36.6	33.2	27.5	19.6	13.9
		MCWB	8.4	10.6	14.9	19.2	20.4	21.7	24.3	22.8	20.2	17.0	12.2	9.4
	10%	DB	10.7	13.7	18.9	25.8	31.9	35.9	37.5	35.4	31.8	25.1	17.4	12.1
		MCWB	7.5	9.5	13.5	18.1	20.1	21.4	23.5	21.9	19.4	15.9	11.3	8.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.1	14.9	18.2	21.6	23.5	25.7	27.5	26.1	23.0	20.2	15.7	12.4
		MCDB	15.3	21.6	25.7	29.6	33.8	36.7	38.3	36.6	34.0	30.1	22.2	16.4
	2%	WB	9.7	12.5	16.5	20.5	22.3	24.1	26.2	24.6	21.7	18.8	13.7	10.9
		MCDB	13.4	17.6	22.1	28.2	32.4	35.8	37.5	35.7	32.7	28.4	20.8	14.1
	5%	WB	8.6	10.9	15.3	19.5	21.5	23.0	25.3	23.7	20.8	17.5	12.7	9.8
		MCDB	11.9	15.3	20.3	26.6	31.4	35.2	36.8	34.8	31.8	26.2	18.8	13.0
	10%	WB	7.5	9.5	14.0	18.6	20.7	22.1	24.3	22.8	20.0	16.3	11.6	8.9
		MCDB	10.3	13.4	18.5	25.2	30.2	33.8	35.8	33.6	30.6	24.2	17.1	11.9
Mean Daily Temperature Range	5% DB	MDBR	8.6	9.8	10.3	11.5	13.3	15.1	15.3	15.8	16.4	13.9	11.2	8.7
		MCDBR	12.4	14.2	14.3	14.9	16.4	17.1	17.1	16.8	18.3	17.1	15.1	13.0
		MCWBR	8.4	8.9	7.9	7.4	7.0	7.2	8.2	7.5	8.8	8.5	8.0	8.3
	5% WB	MCDBR	11.2	13.0	13.1	13.8	14.9	16.3	16.3	16.1	17.9	16.1	13.8	11.1
		MCWBR	7.9	8.6	7.7	7.3	7.3	7.7	8.4	7.8	9.0	8.4	7.8	7.3
Clear-Sky Solar Irradiance	taub	0.403	0.460	0.514	0.525	0.538	0.571	0.555	0.519	0.484	0.513	0.476	0.419	
	taud	2.060	1.927	1.832	1.845	1.838	1.757	1.785	1.857	1.901	1.805	1.883	2.005	
	Ebn at Noon	742	744	745	767	764	738	746	762	762	677	647	685	
	Edh at Noon	124	161	194	203	208	225	217	198	179	179	148	124	
All-Sky Solar Radiation	RadAvg	2.20	2.91	4.27	5.59	6.95	7.95	7.93	7.29	6.13	4.36	2.70	2.00	
	RadStd	0.23	0.36	0.35	0.50	0.44	0.31	0.24	0.19	0.14	0.30	0.29	0.26	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)		N/A	N/A	N/A	N/A	-0.36	-0.68	N/A	N/A	N/A	N/A

Nomenclature: See separate page